

Session 1

Activities

1 Flipping Coins

You have a coin, and you want to determine if it is a fair coin or not. Let's denote the probability that a coin flip results in "H" as p . We want to define an appropriate random variable for the following experiments

1. We flip the coin once.
2. We flip the coin n times.

2 Number of fish in a pond

We want to estimate the total population size N of fish in a pond. In order to do that, we use the capture-recapture method, which consists of the following steps:

- We capture K fish (without replacement).
- We "tag" the captured fish.
- We replace them into the pond.
- We wait for them to mix and we recapture n fish (without replacement).

We are interested in the number of "tagged" fish that we recapture.

1. Define a appropriate random variable for the number of tagged fish that we recaptured.
2. What is the distribution of the random variable?
3. How would you determine the population size N ?

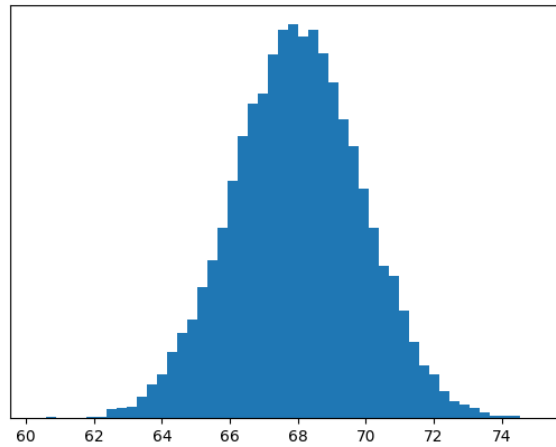
3 Number of visitors at a bakery

We have a bakery we are interested in the number of visitors in a day in order to plan for resources. Let's denote the average number of people that visit the bakery as λ . We observe the bakery's traffic for n days and we record the number of people that visited the website as $X_1, \dots, X_n$.

1. What is a good model for the number of people that visit the bakery in a single day?
2. How would you estimate λ ?
3. What is the probability that the number of visitors is greater 60?

4 Modeling the height of people

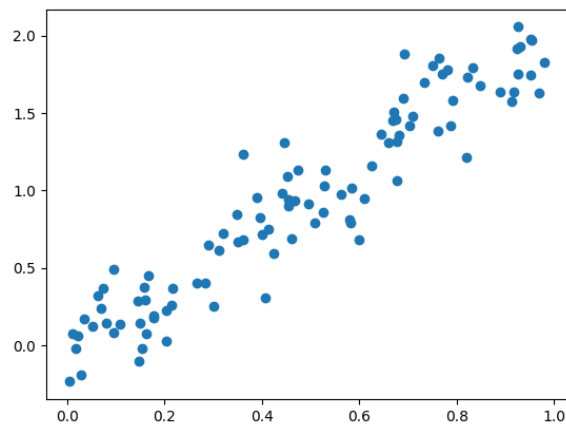
We have a sample of 2500 people and we measure their height.



1. What is a good model for the height distribution?
2. What is the parameter space of the model?
3. What is the law of the random sample?
4. How would we estimate the average height of the population?

5 Regression

We observe data between two variables, X and Y .



1. What is a good model for the relationship between X and Y ?
2. What is the likelihood?
3. How would we estimate the relationship of the variables?